B3.2 Transport

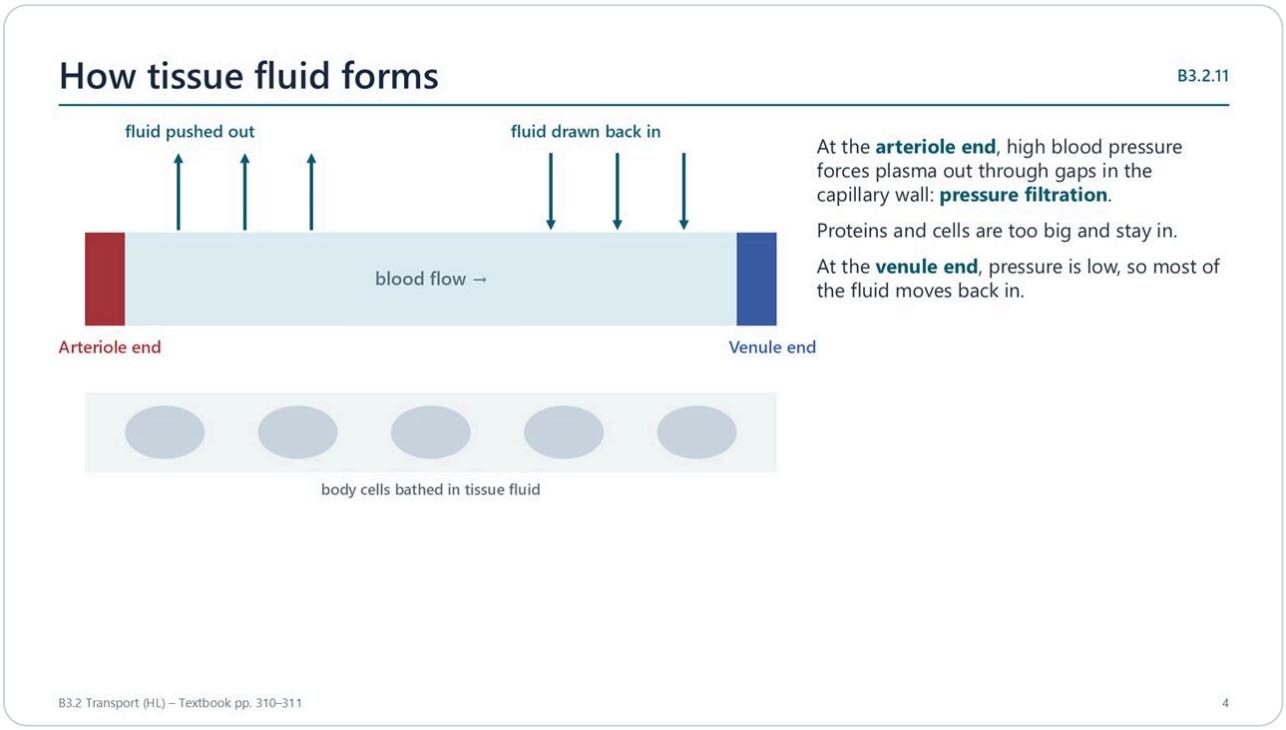
Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-b3-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 310–321 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

TISSUE FLUID AND LYMPH

B3.2.11 Tissue fluid

pp. 310–311



At the arteriole end, high blood pressure forces plasma out through gaps in the capillary wall (**pressure filtration**), leaving proteins and cells behind. At the venule end, blood pressure is lower than the osmotic pull of the plasma proteins, so most of the fluid drains back in.

B3.2.12 Exchange with cells

pp. 311–312

Tissue fluid is plasma without large proteins or red blood cells. Cells take oxygen, glucose, amino acids and ions from it and release carbon dioxide, urea and other wastes into it, by diffusion and membrane transport.

B3.2.13 Lymph

pp. 312–313

Excess tissue fluid enters **lymph capillaries**, which have thin walls with gaps. **Valves** keep lymph flowing one way, and it returns to the blood in veins near the heart. Without drainage, tissues swell (oedema).

B3.2.14 Single and double circulation

pp. 313–314

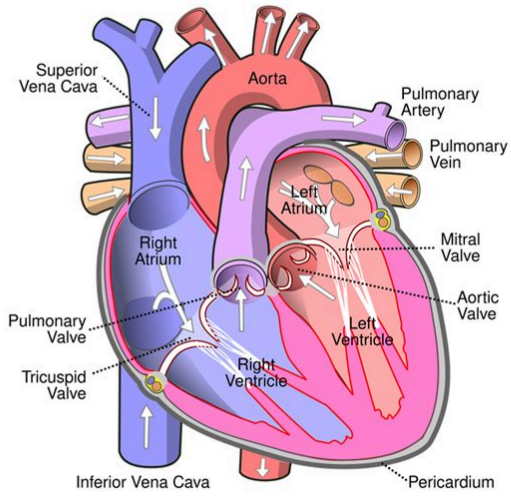
In bony fish, blood goes heart → gills → body → heart, passing through the heart once, and reaches the body at low pressure. Mammals have a pulmonary and a systemic circuit, so blood passes through the heart twice and is re-pumped to the body at high pressure.

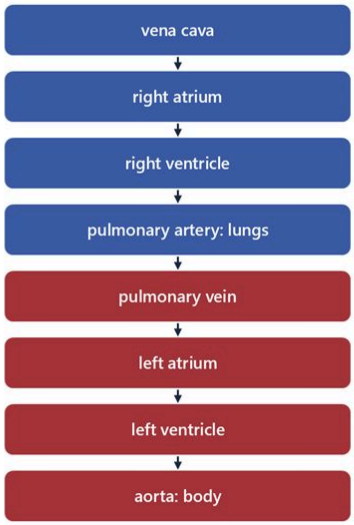
B3.2.15 Adaptations of the heart

pp. 315–318

The heart in frontal section

B3.2.15





Valves between each atrium and ventricle, and at the base of each artery, keep blood moving one way.

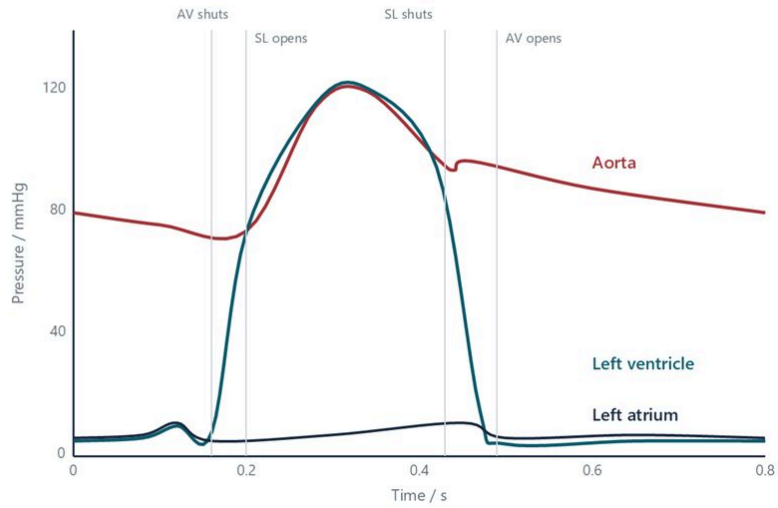
The heart in frontal section. Wapcaplet, CC BY-SA 3.0.

B3.2 Transport (HL) – Textbook pp. 315–318 13

Branched, myogenic **cardiac muscle** contracts rhythmically; the **SA node** sets the pace. Thin-walled **atria** fill the thick-walled **ventricles** (left thickest). **AV** and **semilunar valves** stop backflow, the **septum** separates the two sides and **coronary vessels** supply the muscle.

Pressures in the left side of the heart

B3.2.16



Schematic, one beat at 75 beats per minute. AV: atrioventricular valve. SL: semilunar valve.

Valves open and close because of **pressure differences**: a valve opens when the pressure behind it is higher than in front.

Systole: contraction. **Diastole**: relaxation.

B3.2 Transport (HL) – Textbook pp. 318–321 19

The SA node fires, the atria contract, the AV node delays the impulse, then the ventricles contract. Valves open and close with pressure differences: AV valves shut as ventricular pressure rises; semilunar valves open when it exceeds arterial pressure and shut when the ventricles relax. Blood pressure is written systolic/diastolic, for example 120/80 mmHg.

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